Statistics 2 2026/27 Exam

Introduction

The dataset `amsterdam-house-prices.csv` contains information on the asking price of a random sample of houses for sale in Amsterdam in August 2021, as well as some basic information on the listings.

The definitions of all the variables are as follows:

- `price`: The asking price in thousands of euros.
- `area`: The floor space in square meters (m^2).
- `bedrooms`: The number of bedrooms.
- `neighborhood`: Whether the listing was in the North, South, East, West, or Central neighborhood of Amsterdam.

Use this dataset to answer the following questions.

Question 1

What is the sample correlation between `price` and `area`?

Model 1

Estimate a linear regression model with `price` as the dependent variable and `area` as the independent variable.

If you estimated the model correctly, you should have an estimated intercept of 23.3385.

Question 2

What is the sample regression slope?

Question 3

According to the sample regression slope, a $1m^2$ increase in the area of a house on average increases the listing price by how many euros?

Note: Pay attention to the units of the variables. Price is measured in *thousands* of euros. You will need to multiply by 1000 to report your answer in euros. Pay attention to the units in all questions that follow.

Question 4

Plot the residuals (on the vertical axis) against `area` (on the horizontal axis). Choose the answer below which best describes what you see:

- The dispersion of the residuals increases as area increases.
- The dispersion of the residuals decreases as area increases.
- The dispersion of the residuals is constant in area.
- The residuals are all positive.
- The residuals are all negative.

Based on this, which model assumption is likely violated?

- Linear in parameters
- Random sampling
- Sample variation in the explanatory variable
- Zero conditional mean
- Homoskedasticity
- Normality

Model 2

Estimate a linear regression model with `price` as the dependent variable and `bedrooms` as the independent variable.

If you estimated the model correctly, you should have an estimated intercept of 166.604.

Question 5

The estimated intercept of 166.604 can be interpreted as follows.

The expected price of houses with _____ bedrooms is €_____. However, this estimate is unreliable because:

- This number of bedrooms is outside the range of our data.
- The estimated intercept is not statistically different from zero.
- The estimated slope is not statistically different from zero.
- The adjusted R^2 is smaller than the standard R^2.

Question 6

Perform a formal test for heteroskedasticity by regressing the square of the residuals on `bedrooms`.

Use a 5% significance level.

In this test, the null hypothesis is that the model exhibits which of the following:

- Homoskedasticity
- Heteroskedasticity

- Multiskedasticity
- Paraskedasticity

What is the p-value from this test? _____

What is the conclusion? Choose one of the answers below:

- The model exhibits heteroskedasticity. Therefore the standard errors are not reliable.
- The model exhibits homoskedasticity. Therefore the standard errors are not reliable.
- The model exhibits heteroskedasticity. Therefore the standard errors are reliable.
- The model exhibits homoskedasticity. Therefore the standard errors are reliable.

Model 3

Estimate a regression model with `price` as the dependent variable and the following 2 independent variables:

1. `area`
2. `bedrooms`

If you estimated the model correctly, your estimated intercept should equal 92.531.

Question 7

What proportion of the variation in `price` is left unexplained by the model?

Note: your answer should be between 0 and 1.

Question 8

Use the model to test the following claim at the 5% level:

"An additional bedroom in a listing on average decreases the asking price by more than €40,000, holding the area fixed."

Perform this test by answering the questions below. Use the conventional standard errors given by `summary()`.

- What is the null hypothesis? $\beta_2 < / \leq / > / \geq / = / \neq$ _____ (choose one comparison operator and fill in a value in the blank).
- What is the alternative hypothesis? $\beta_2 < / \leq / > / \geq / = / \neq$ _____ (choose one comparison operator and fill in a value in the blank).

- Under the null hypothesis, the test statistic $T = (B_2 - b)/S_{B_2}$, where b is the hinge, follows a t distribution with how many degrees of freedom? _____
- What is the value of the test statistic? _____
- What is the associated p-value? _____
- What is your conclusion? Choose an option below:
  - Reject H0: There is sufficient evidence for the claim.
  - Reject H0: There is not sufficient evidence for the claim.
  - Don't reject H0: There is sufficient evidence for the claim.
  - Don't reject H0: There is not sufficient evidence for the claim.

Question 9

Use the estimated model to obtain a 95% confidence interval for the average price of 3-bedroom listings that are 80m^2 in size. Report your answer in euros.

Model 4

Estimate a regression model with `price` as the dependent variable and the following 4 independent variables:

1. A dummy variable that equals 1 if the listing is in the East and 0 otherwise.
2. A dummy variable that equals 1 if the listing is in the North and 0 otherwise.
3. A dummy variable that equals 1 if the listing is in the South and 0 otherwise.
4. A dummy variable that equals 1 if the listing is in the West and 0 otherwise.

Note: By using the variable `neighborhood` as is in your model formula, these dummy variables will be created automatically.

If you estimated the model correctly, your estimated intercept should equal 719.41.

Question 10

According to the estimated model, what is the average price of listings in the central neighborhood?

Note: Report your answer in euros, taking note of the units of your variables.

Question 11

Let $\bar{p}_j$ be the average price (in €1,000s) of a listing in neighborhood j, where j is one of Central, East, North, South or West.

The estimated coefficient in front of the East neighborhood dummy in the model is -269.12. Which of the following does this number represent?

- $\bar{p}_{East}$

- $\bar{p}_{East} - \bar{p}_{Central}$
- $\bar{p}_{Central} - \bar{p}_{East}$
- $\bar{p}_{Central} + \bar{p}_{East}$
- $\bar{p}_{Central}$

Question 12

Test the following claim at the 5% level: "There are differences in average house prices across neighborhoods".

- The null hypothesis is that *all/at least one/none* of $\beta_j =$ _____ for j from _____ to _____.
- The alternative hypothesis is that *all/at least one/none* of $\beta_j \neq$ _____ for the same j.

The test statistic is of the form:

$$\frac{\frac{a}{k}}{\frac{SSE}{n-k-1}}$$

What is the numeric value of a in the test? _____

What is the numeric value of the test statistic? _____

What is the critical value? _____

Which of the 4 options below is the correct conclusion from the test?

- Reject H0. There are significant differences in house prices across neighborhoods.
- Reject H0. There are no significant differences in house prices across neighborhoods.
- Don't reject H0. There are significant differences in house prices across neighborhoods.
- Don't reject H0. There are no significant differences in house prices across neighborhoods.

Model 5

Estimate a regression model with `price` as the dependent variable and the following 3 independent variables:

1. `area`
2. `bedrooms`
3. The interaction between `area` and `bedrooms`.

If you estimated the model correctly, your estimated intercept should equal 169.7672.

Question 13

Which variables are individually statistically significant at the 5% level?

Question 14

Obtain a 95% confidence interval for the coefficient in front of the variable `bedrooms`. Report the lower and upper bound.

Question 15

According to the model estimates, for a one-bedroom listing, by how much on average does the listing price increase (in €) when the area increases by 1m^2?

Question 16

According to the model estimates, for a 100m^2 listing, by how much on average does the listing price change (in €) when it has one more bedroom (holding area fixed at 100m^2)?